RoboValue

Simulation — 15 tasks

Real World — 20 tasks

Expert Demostrations + Failure and Recovery + Long-Horizon Temporal + Multi-Solution

Can Value Models Reliably Understand Task State and Instructions?

Correct

Hang all three mugs on the mug rack.

Counterfactual

- Hang all three **plates** on the mug rack.
- **Stand** all three mugs **upright** on the table.
- Place all three mugs **beside** the mug rack.
- Hang exactly **two** mugs on the mug rack and leave the **remaining** mug on the table.

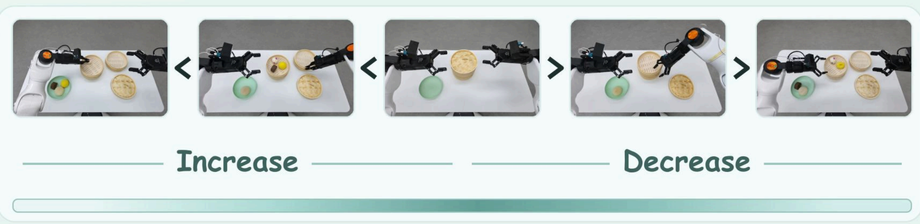


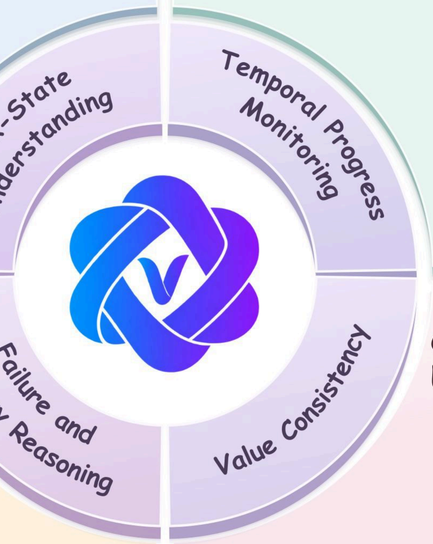
v(A) > v(B) v(Correct) > v(Counterfactual)

Metrics
- SA
- TGA
- SIA

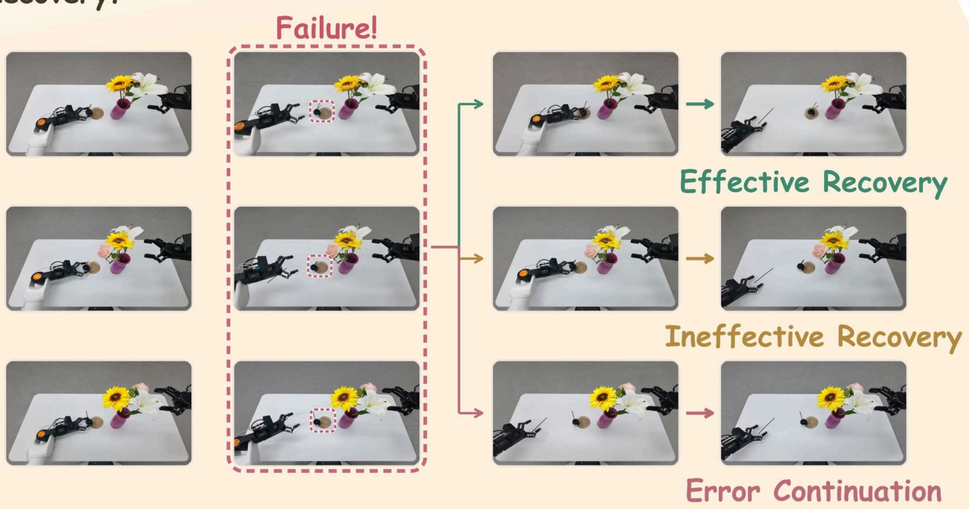
Can Value Models Accurately Track Execution Progress?

Metrics
- VOC
- Cycle-VOC
- Memory-VOC





Can Value Models Diagnose Failures and Recognize Effective Recovery?



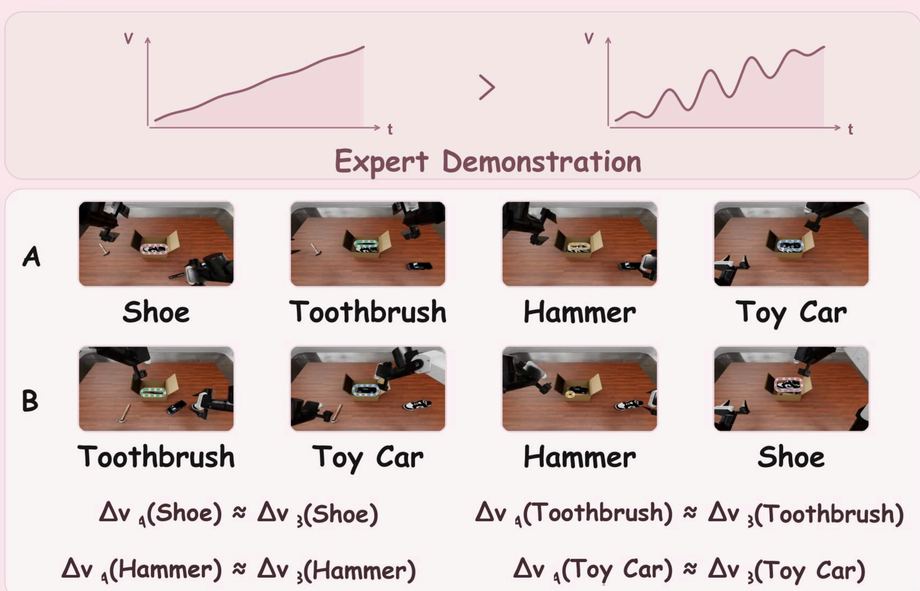
v(Effective Recovery) > v(Ineffective Recovery)

v(Effective Recovery) > v(Error Continuation)

Metrics
- FPL
- TRR

Metrics
- VS
- CSVC

Can Value Models Assign Semantically Consistent Values across Valid Executions?



$\Delta v_4(\text{Shoe}) \approx \Delta v_3(\text{Shoe})$ $\Delta v_4(\text{Toothbrush}) \approx \Delta v_3(\text{Toothbrush})$

$\Delta v_4(\text{Hammer}) \approx \Delta v_3(\text{Hammer})$ $\Delta v_4(\text{Toy Car}) \approx \Delta v_3(\text{Toy Car})$

Unified Interfaces

Scalar Value	Pairwise Comparison	Subtask Description
v(A; I)	c(A1, A2; I)	s(A; I)

A: query I: instruction

Preserve Native Value Semantics

Open-Source

- Training Dataset
- Codes & Tutorials
- Leaderboard
- Evaluation Support
- Diverse Value Models
- Community